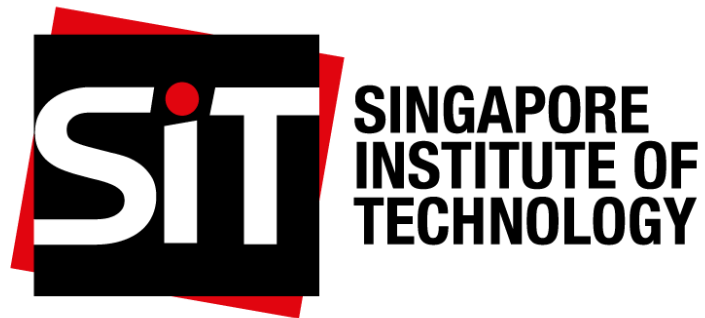




ICT2113: Software Modelling and Analysis
[AY 2024/2025]

Eliciting PS
[Project Team P1 - 1039T]

Group Members:

Tan Tze Han	2301483
Jolie Ngai Ning Li	2403175
Koh Tong Wei	2402162
Tho Jun En Thaeus	2401039
Yap Hong Xiang Alfred	2401438

30 Jan 2026

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1. Customer Company Description

1.1 Company Profile

Name : Great Western Financial Consultancy

Address : 12 New Punggol Rd, Singapore 828616

Commercial Sector : Financial Services

Line of Business : Financial Consultancy and Advisory Services

1.2 Company History and Capabilities

Great Western is a respected financial consultancy firm operating in Singapore's financial services sector. With a proven track record in financial management systems and process optimization, Great Western has demonstrated expertise in modernizing financial workflows for diverse clients.

In 2019, Great Western successfully led a financial workflow digitalization project for a mid-sized retail client, implementing automated budgeting, expense tracking, and monthly financial reporting systems. This initiative achieved remarkable results, reducing manual data entry errors and improving reporting turnaround time by over 30%, demonstrating their commitment to leveraging technology for operational excellence.

Mission : To empower individuals and businesses to achieve financial wellness through expert guidance, innovative tools, and personalized strategies that transform their relationship with money.

Vision : To be Singapore's premier financial consultancy recognized for combining traditional advisory excellence with cutting-edge digital solutions that deliver measurable, lasting financial outcomes for our clients.

Core Business : Great Western provides comprehensive financial consultancy services like personal financial planning and wealth management, business financial strategy and optimization and investment advisory and portfolio management.

Industry Sector : Financial Services, Financial Consultancy and Advisory

Key Initiatives related to new software :

Great Western clients require practical, accessible tools to implement and maintain the financial strategies we recommend between consultations. Currently, many clients struggle with consistent expense tracking, budget adherence, and financial awareness, which limits the effectiveness of our guidance and reduces their financial outcomes. This strategic initiative aims to provide our clients with a professional mobile application that complements our consultancy services with

daily financial management capabilities as well as empower our clients to take ownership of their financial health and provide real time financial visibility and budget control. It should also generate data insights that can enhance our consultancy effectiveness and position Great Western as an innovative, technologically advanced financial services provider.

2. Project Description

2.1. Project Name

We decided to name our project :

“Money Management Application for Personal Expense Planning and Budget Control”

2.2. Project Context :

The modern financial landscape has been transformed by mobile technology adoption. Since November 2016, mobile devices have generated 48.19% of global network traffic. This mobile first reality presents both opportunities and challenges for personal financial management.

Despite widespread smartphone adoption, individuals face increasing complexity in managing their finances. For example, individuals may experience multiple income streams from diverse sources, proliferation of digital payment methods and subscription services, the lack of real time financial visibility and limited tools for proactive budget management.

Traditional Personal finance management approaches such as spreadsheets, paper ledgers or banking apps fail to provide comprehensive, intuitive, real time financial oversight that modern users require. While numerous mobile finance applications exist in the market, they typically have limitations such as platform exclusivity (iOS only or android only), complex interfaces that discourage regular use, lack of integrated budget monitoring and alerts and the absence of meaningful visual analytics for spending patterns.

Technical Context :

The challenge of reaching the maximum number of users requires supporting both Android and iOS platforms, which traditionally demands developing and maintaining two separate codebases using different programming languages (Java/Kotlin for Android, Swift for iOS). This dual development approach significantly increases development costs, extends timelines, and creates potential inconsistencies in user experience across platforms.

The flutter framework which was developed and maintained by Google, addresses this challenge by enabling truly cross platform development using a single Dart codebase. This approach delivers native level performance on both platforms while reducing development time and costs by approximately 40 to 50% compared to traditional dual platform development.

2.3. Project Goals

Goal 1 : Comprehensive Financial Transaction Management

- **Specific:** Develop a mobile application enabling users to record, categorize, track, update and delete both income and expense transactions with comprehensive category level organization and transaction history maintenance
- **Measurable:** Support minimum 15 pre-defined expense categories and 5 income categories, with the ability to record unlimited transactions
- **Attainable:** Utilize Flutter's robust UI framework and local database capabilities to implement comprehensive data entry and retrieval features
- **Relevant:** Directly addresses Great Western clients' need for organized financial record keeping to support consultancy recommendations
- **Time-bound:** Core tracking functionality to be completed by Month 2 of development cycle

Goal 2: Visual Financial Analytics

- **Specific:** Provide graphical representations of financial data enabling users to understand spending patterns and trends
- **Measurable:** Implement minimum 3 chart types (monthly expense breakdown, annual trends, category-wise analysis) with monthly and annual view options
- **Attainable:** Leverage Flutter's charting libraries and data visualization packages
- **Relevant:** Supports informed decision making aligned with Great Western's consultancy approach of data driven financial planning
- **Time-bound:** Visualization features to be implemented by Month 3 of development cycle

Goal 3: Proactive Budget Management

- **Specific:** Implement budget setting functionality with automated warnings when spending approaches or exceeds defined limits
- **Measurable:** Enable users to set monthly budgets with real time tracking and automatic notifications at 80% and 100% of budget thresholds
- **Attainable:** Use Flutter's local notification services and real time calculation capabilities
- **Relevant:** Directly supports budget adherence, a critical component of effective financial management
- **Time-bound:** Budget management features to be completed by Month 3 of development cycle

3. Business Operations Requirements (3 pages)

3.1. Level 0 : Personal Financial Management System

BOR-L0-01: Personal Financial Management System

Precedence: High

The business shall provide personal financial management operations that enable individuals to record, monitor, and analyse income, expenses and budget in order to support informed financial planning and consultancy-driven decisions.

Constraints and Business Rules:

- The business shall operate the solution as a non-transactional financial management service.
- The business shall not execute monetary payment or transfers on behalf of users.
- The business shall rely on user-provided financial data for analyses and insights.

3.2. Level 1 : Sub-requirements of Primary Business Operations

BOR-L1-01: Expense Management

Precedence: High

The business shall support structured recording of personal expenses to enable spending visibility and comparison against budget.

Constraints and Business Rules:

- The business shall require expenses to be classified using predefined categories.
- The business shall require each expense record to include an amount, category, and transaction date.
- The business shall allow expense records to be updated or deleted to correct inaccuracies.
- The business shall support the recording of recurring expense entries for regularly occurring monthly expenses.

BOR-L1-02: Income Management

Precedence: High

The business shall support the management of multiple personal income sources to establish an accurate financial baseline for users.

Constraints and Business Rules:

- The business shall require each income record to include an amount, category and date.
- The business shall allow income records to be amended or removed to correct inaccuracies.
- The business shall support the recording of recurring income entries for regularly occurring monthly expenses.

BOR-L1-03: Budget Planning and Monitoring

Precedence: Medium-High

The business shall support proactive budget planning and monitoring by enabling users to define month budget limits and track expenses against the limits.

Constraints and Business Rules:

- The business shall support the recording of a monthly budget.
- The business shall continuously evaluate expenses relative to defined budgets.
- The business shall not restrict expense recording when budget limits are exceeded

BOR-L1-04: Financial Reporting and Analytics

Precedence: Medium

The business shall provide financial summaries and analytical views that enable users to understand spending patterns and financial trends.

Constraints and Business Rules:

- The business shall generate reports solely recorded income and expenses transactions.
- The business shall support both monthly, quarterly and annual reporting periods.
- The business shall present reports in a read-only manner without altering present data.

BOR-L1-05: User Notification Operations

Precedence: Medium

Requirement Clause: The business shall provide advisory notifications to alert users when spending approaches or exceeds defined budget thresholds.

Constraints and Business Rules:

- The business shall issue notifications at predefined budget thresholds levels.
- The business shall ensure notifications are informational.
- The business shall base notification triggers on a monthly budget calculation.

4. Technical Requirements

4.1. Level 0 : Primary Technical Requirement

Tech-L0-01: Cross-Platform Mobile Application Using Flutter

Precedence: Essential

The application shall be implemented as a cross-platform mobile application for Android and iOS using the Flutter framework, with secure local and cloud data storage and the ability to operate in an offline-first manner.

Tech-L0-02: Data Security, Privacy, and Recovery

Precedence: Essential

The application shall protect users' data through industry standard security measures, comply with Singapore data protection regulations, and provide mechanisms for backup and recovery to prevent data loss in the event of device failure or replacement.

Tech-L0-03: Backend Services and External Integration

Precedence: Essential

The application shall include a modular backend architecture to support authentication, data synchronisation, and administrative functions, while maintaining the ability to integrate with external systems and services without disrupting core application functionality.

Tech-L0-04: Development Standards and Maintainability

Precedence: Essential

The application shall be developed using agreed technologies and coding standards, with proper configuration management, documentation, and development infrastructure to ensure long term maintainability, support, and the ability to deliver updates and enhancements efficiently.

4.2. Level 1 : Technical Sub-Requirements

Under Tech-L0-01: Cross-Platform Mobile Application Using Flutter

Tech-L1-01: Supported Platforms and Versions

Precedence: Essential

The application shall support recent Android and iOS versions commonly in use at the time of deployment, enabling at least the last two major OS versions on each platform, subject to store policies. The app shall be publishable on both Google Play Store and Apple App Store.

Tech-L1-02: Local Data Storage and Offline-First Behaviour

Precedence: Essential

The application shall function in an offline-first manner, allowing users to record and view transactions, budgets, and charts without a continuous internet connection via secure local storage on the device. Data created offline shall remain on the device and appear in all relevant summaries. Any future synchronisation capability shall not break offline usability.

Tech-L1-03: Performance and Responsiveness**Precedence: Essential**

The application shall provide a responsive user experience, with typical screens (dashboard, budget overview) loading and rendering within a reasonable time on mid-range devices. For planning purposes, it expects key screens to render within approximately two seconds under normal data volumes.

Tech-L1-04: Implementation Technologies**Precedence: Essential**

The application shall be implemented using Dart within the Flutter framework, and any back-end services, if used, shall be built with widely supported languages and frameworks agreed between Great Western and Pear Developer inc (e.g., .NET, Node.js, or equivalent) to ensure maintainability and availability of future development resources.

Under Tech-L0-02: Data Security, Privacy, and Recovery**Tech-L1-05: Data Security and Privacy****Precedence: Essential**

The application shall protect users' data through appropriate security measures such as device-level protection, secure storage mechanisms, and adherence to applicable data protection regulations in Singapore. The app shall support optional app level access control (e.g., PIN, biometric unlock) where supported by the device. Personal data shall not be shared with third parties without explicit user consent.

Tech-L1-06: Backup and Recovery Mechanism**Precedence: Desirable**

The application shall support a mechanism for users to back up and restore their data (e.g., via export and re-import, or via platform-standard backup facilities), so that accidental loss of device does not automatically imply permanent data loss. Any backup feature shall clearly inform users about what is backed up and where it is stored.

Under Tech-L0-03: Backend Services and External Integration**Tech-L1-07: Lightweight, Service-Oriented Back-End****Precedence: Essential**

The application backend server shall follow a modular, service-oriented architecture that cleanly separates APIs for authentication, data synchronisation, and administrative reporting, so that changes to one service do not disrupt the others.

Tech-L1-08: Integration Readiness**Precedence: Acceptable**

The application shall be designed to allow future integration with Great Western's advisory systems or external services (payment gateways, etc) without requiring a complete re-implementation of core functionality. Public APIs or data formats used for export shall be reasonably stable or versioned.

Under Tech-L0-04: Development Standards and Maintainability

Tech-L1-09: Maintainability and Support Window

Precedence: Essential

The application shall be designed to support corrective maintenance and minor enhancements without disrupting existing user data. Code, configuration, and deployment processes shall be structured so that defects can be addressed through app updates within an agreed warranty period after launch.

Tech-L1-10: Administration and Configuration

Precedence: Desirable

The application shall provide basic configuration options (e.g., currency format, default period view, default notification settings) that users can adjust within the app's settings screen. Default settings shall be defined by Great Western to reflect its recommended usage patterns.

Tech-L1-11: Development Workstation Requirements

Precedence: Essential

Development workstations shall meet the following minimum specifications:

- Operating System: Windows 7 or higher, or macOS Sierra and above (for iOS builds)
- Processor: Intel Core i3 or equivalent processor or higher
- RAM: 8 GB or higher (16 GB recommended for optimal performance)
- Storage: 100 GB available disk space or higher (SSD recommended)

Testing Devices: Android phones running Android 6.0 and above, and iPhones running iOS 9 and above, for physical device testing.

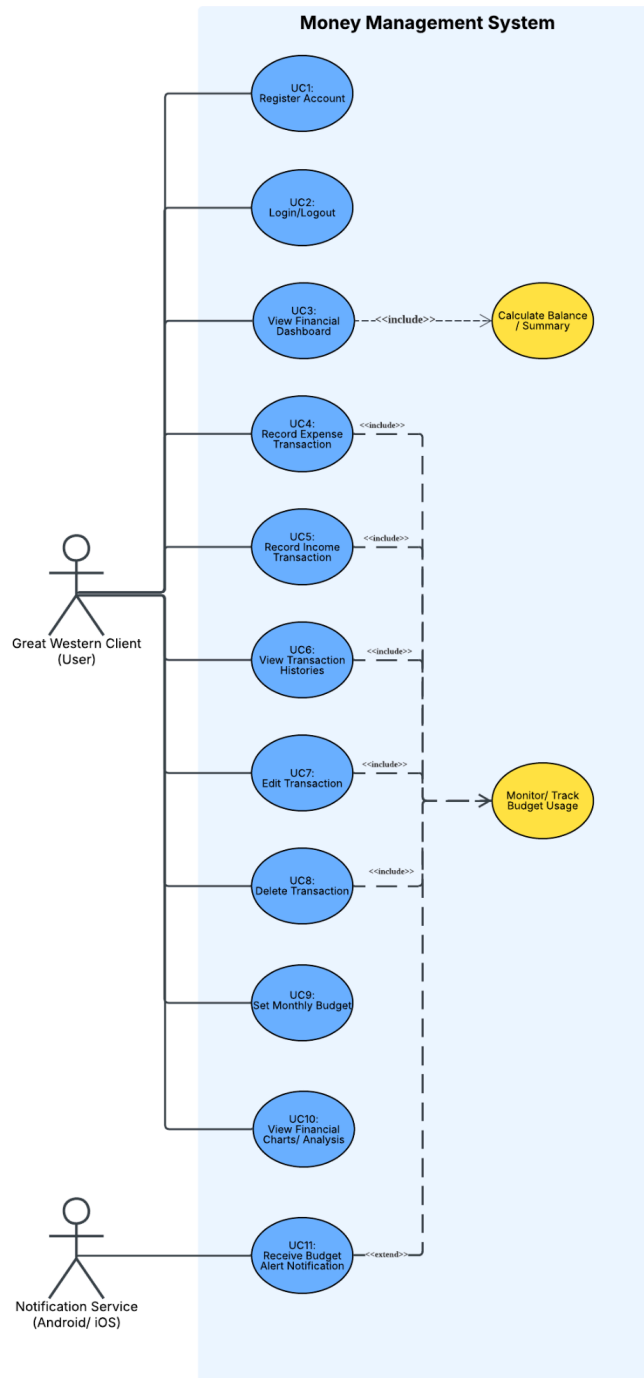


Fig 1 : Use Case Diagram

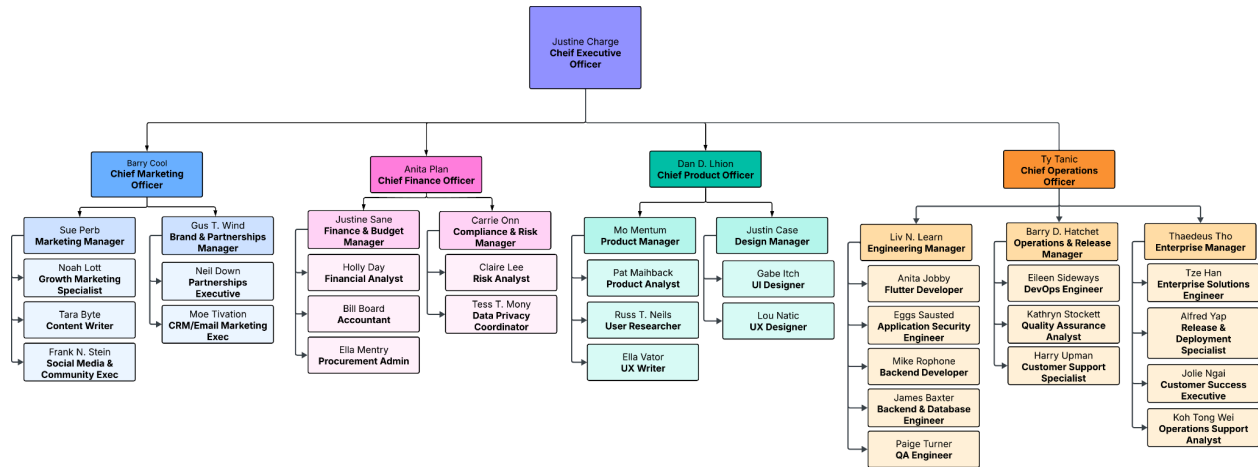


Fig 2 : Organization Chart

5. Financial Requirements

5.1. Project Budget Breakdown

Allocated Budget: \$50, 000

Used Budget: \$43,00

Department	Project Stage	Description	Estimated Costs (\$)
Product Department (CPO)	Requirements & Planning	Scope definition, requirements workshop, user stories, technical plan, project plan formation	3,000
Product Department (CPO)	UI/UX Design	Wireframes, prototype, user flow validation	5,000
Operations Department (COO)	Flutter App Development	Core tracking (Month 2), analytics charts, budget logic and notifications (Month 3)	18,000
Operations Department (COO)	Backend / Local Database Setup	Local storage (e.g., SQLite), data models, CRUD, sync-ready architecture	6,000
Finance Department (CFO)	Cybersecurity Development	Protect users' financial data with secure authentication and encrypted local storage	5,000
Operations Department (COO)	Testing & QA	test cases, bug fixing cycles, cross-device testing (Android + iOS)	4,000
Marketing (CMO)	Marketing & Launch	App branding assets, launch plan, basic promo content	2,000

5.2. Payment Schedule

Milestone	Time	% of Allocated Costs	Costs (\$)
M1	Month 1 (Week 1–2)	20%	8,600
M2	Month 2 (Week 3–6)	30%	12,900
M3	Month 2 (Week 7–9)	15%	6,450
M4	Month 3 (Week 10–12)	20%	8,600
M5	Month 4 (Week 13–16)	15%	6,450

6. Delivery Requirements and Project Timeline

Project Duration: 4 months

The project consists of 3 months development and 1 month of testing, which will then finalize and prepare the application for deployment.

Milestone	Timeline	Objective	Deliverables
M1	Month 1 (week 1-2)	Project Initiation & Requirements Approval	Approved project scope, validated business operations requirements, and confirmed project plan
M2	Month 2 (week 3-6)	Core Financial Transaction Management	Income and expenses management features, category-based recording, update and deletion of transactions, and transaction history management
M3	Month 2 (week 7-9)	Visual Financial Analytics	Vivid graphical view of financial summaries of monthly, quarterly, and annual analysis
M4	Month 3 (week 10-12)	Proactive Budget Management	Monthly budget setup, expense tracking against budgets, and automated warning notifications
M5	Month 4 (week 13-16)	System Testing, Final Review & Handover	Complete system testing, defects resolved, documentation, and deployment of final application

Project Management Approach: Agile

Progress reviews to be conducted every sprint lasting 2-3 weeks, which is during the completion of each milestone, and 1 more in the middle of M5, to assess delivery status, validate completed functionality, and identify any risks or required adjustments. Each progress report should include:

- Milestone completion status
- Summary of delivered features
- Identified issues and mitigation actions
- Confirmation of readiness to proceed to the next milestone

This approach ensures that Great Western Financial Consultancy maintains clear visibility over project progress while allowing controlled and timely delivery of system capabilities.

7. Sign Off

This Project Specification (PS) has been prepared by Great Western Financial Consultancy for the purpose of soliciting proposals from qualified software development vendors for the Money Management Mobile Application project.

Date Prepared: January 29, 2026

Prepared By: Great Western Project Lead Team

Great Western Approvals :

Chief Executive Officer (CEO) : **Justine Charge**

Date: 30 January 2026

Chief Marketing Officer (CMO) : **Barry Cool**

Date: 30 January 2026

Chief Finance Officer (CFO) : **Anita Plan**

Date: 30 January 2026

Chief Product Officer (CPO) : **Dan D. Lhion**

Date: 30 January 2026

Chief Operations Officer (COO) : **Ty Tanic**

Date : 30 January 2026

8. Addendum and attachment

8.1 Company Letterhead



8.2 Technical Requirements Development

This section documents the rationale, thought process, and decomposition approach used to develop the technical requirements in Section 4

8.2.1 Requirements Tree Decomposition Approach

The technical requirements were developed using a top-down Requirements Tree methodology to ensure completeness and logical organization. The process followed these steps:

1. Identify Level 0:
 - a. We started by identifying the broad technical capabilities needed to deliver the business goals. Four major themes emerged:
 - Cross-platform mobile application delivery
 - Data security and user privacy protection
 - Backend infrastructure and integration capability
 - Long-term maintainability and development standards
2. Decompose into Level 1
 - a. Each Level 0 requirement was systematically broken down into specific, measurable, and testable technical sub requirements by asking:

- What specific technical capabilities does this theme require?
 - What constraints or standards must be met?
 - What are the customer's non-negotiable versus desirable technical expectations?
3. Applying the Magnificent Seven Quality Criteria:
- a. Each Level 1 requirement was reviewed to ensure it met the seven quality criteria

8.2.2 Design Decisions and Trade-offs

Decision 1: Flutter vs. Native Development

Options considered: Native iOS (Swift) + Native Android (Kotlin) vs. Flutter vs. React Native

Choice: Flutter framework

Rationale:

- Single codebase reduces development time by ~40-50% compared to dual native
- Near-native performance suitable for financial app requirements
- Pear Developer Inc has established Flutter expertise

Trade-off: Slightly larger app size vs. significantly lower development and maintenance costs

Decision 2: Offline-First vs. Cloud-First Architecture

Options considered: Cloud-first (requires internet) vs. Offline-first (local storage primary)

Choice: Offline-first with cloud sync

Rationale:

- Users need to record transactions anytime, anywhere (even without connectivity)
- Financial data privacy concerns favor local storage
- Reduces backend infrastructure complexity and costs for initial release

Trade-off: More complex data synchronization logic vs. better user experience and privacy

8.2.3 Technical Constraints and Assumptions

Constraints:

- Must comply with Google Play Store and Apple App Store policies
- Must adhere to Singapore Personal Data Protection Act (PDPA)
- Timeline requires delivery within 3-month core implementation window

Assumptions:

- Target users have smartphones running OS versions from last 2-3 years
- Typical user will store 1,000-5,000 transactions over app lifetime
- Internet connectivity available for cloud features

8.3 Business Operations Requirements Development

This section documents the rationale, thought process, and decomposition approach used to develop the Business Operations Requirements presented in Section 3.

8.3.1 Requirements Tree Decomposition Approach

The Business Operations Requirements were developed to ensure business intent was clearly articulated, logically structured, and traceable to the project goals and customer profile.

Identify Level 0

Identifying the main business function that the solution would support was the first step in the development process. The overall company's ability has been determined to be: Personal financial transaction management for individual users based on the project objectives and Great Western's consulting profile.

Decompose into Level 1

The Level 0 business operation was systematically decomposed into Level 1 primary business operations by examining the project goals. Through this process level 1 business operations were identified.

- Expense management
- Income management
- Budget planning and monitoring
- Financial reporting and analytics
- User notification operations

8.3.1 Business Constraints and Assumptions

Constraints

- The business shall operate as an advisory and planning support tool.
- Financial data shall be entered and maintained by the user.
- Expenses and income category shall be predefined to ensure consistency.
- The budget shall be defined on a monthly basis.

Assumptions

- The solution is used to support personal financial planning and consultancy discussions.
- It is assumed that the application will not support or execute any financial transactions such as payments or transfer.
- It is assumed that each set of financial data is associated with a single user and is not shared across multiple users.

8.4 Initial Table of Requirements

Technical Requirements	Codes
Cross-Platform Mobile Application Using Flutter	T-L0-1
Data Security, Privacy, and Recovery	T-L0-2
Backend Services and External Integration	T-L0-3
Development Standards and Maintainability	T-L0-4
Supported Platforms and Versions	T-L1-1
Local Data Storage and Offline-First Behaviour	T-L1-2
Performance and Responsiveness	T-L1-3
Implementation Technologies	T-L1-4
Data Security and Privacy	T-L1-5
Backup and Recovery Mechanism	T-L1-6
Lightweight, Service-Oriented Back-End	T-L1-7
Integration Readiness	T-L1-8
Maintainability and Support Window	T-L1-9
Administration and Configuration	T-L1-10
Development Work Station Requirement	T-L1-11
Business Requirement	
Personal Financial Management System	B-L0-1
Expense Management	B-L1-1
Income Management	B-L1-2
Budget Planning and Monitoring	B-L1-3
Financial Reporting and Analytics	B-L1-4

User Notification Operations	B-L1-5
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8.5 Organization Chart Role Distribution

Total Employee Count: 41

Division	Role	Name	Primary Responsibilities
Executive	Chief Executive Officer (CEO)	Justine Charge	Provides overall strategic leadership, approves major decisions, and oversees delivery across all departments.
Marketing	Chief Marketing Officer (CMO)	Barry Cool	Defines marketing strategy, brand positioning, and growth direction aligned with business goals.
Marketing	Marketing Manager	Sue Perb	Plans and coordinates marketing campaigns, schedules activities, and tracks performance metrics.
Marketing	Growth Marketing Specialist	Noah Lott	Drives user acquisition and retention through growth experiments and optimisation of funnels.
Marketing	Content Writer	Tara Byte	Produces marketing copy, onboarding content, and app store descriptions.
Marketing	Social Media & Community Executive	Frank N. Stein	Manages social media presence, community engagement, and user feedback channels.
Marketing	Brand & Partnerships Manager	Gus T. Wind	Develops brand strategy and manages external partnerships and collaborations.
Marketing	Partnerships Executive	Neil Down	Coordinates partner outreach, onboarding, and partnership deliverables.
Marketing	CRM / Email Marketing Executive	Moe Tivation	Manages email campaigns and CRM workflows to improve user engagement and retention.
Finance	Chief Finance Officer (CFO)	Anita Plan	Oversees budgeting, financial planning, and financial risk management for the project.

Finance	Finance & Budget Manager	Justine Sane	Tracks project spending, manages budgets, and prepares financial reports.
Finance	Financial Analyst	Holly Day	Analyses project costs, forecasts spending, and identifies financial risks.
Finance	Accountant	Bill Board	Handles accounting records, expense tracking, and financial documentation.
Finance	Procurement Administrator	Ella Mentry	Manages procurement requests, vendor documentation, and purchasing records.
Finance / Risk	Compliance & Risk Manager	Carrie Onn	Oversees compliance, risk management, and security governance requirements.
Finance / Risk	Risk Analyst	Claire Lee	Identifies and monitors project risks, proposing mitigation strategies.
Finance / Risk	Data Privacy Coordinator	Tess T. Mony	Ensures user data handling complies with privacy and protection requirements.
Product	Chief Product Officer (CPO)	Dan D. Lhion	Defines product vision and ensures alignment between user needs and business goals.
Product	Product Manager	Mo Mentum	Owns product roadmap, prioritises features, and coordinates stakeholder expectations.
Product	Product Analyst	Pat Maihback	Supports product decisions using data analysis and feature performance metrics.
Product	User Researcher	Russ T. Neils	Conducts user research and usability testing to validate requirements and designs.
Product	UX Writer	Ella Vator	Writes in-app text and onboarding content to improve clarity and usability.
Design	Design Manager	Justin Case	Leads the UI/UX team and ensures consistent design standards across the application.
Design	UI Designer	Gabe Itch	Designs visual interfaces and high-fidelity screens following brand

			guidelines.
Design	UX Designer	Lou Natic	Creates wireframes, user flows, and prototypes to optimise user experience.
Operations	Chief Operations Officer (COO)	Ty Tanic	Oversees operational execution, delivery timelines, and system readiness.
Operations / Engineering	Engineering Manager	Liv N. Learn	Manages the engineering team and ensures timely, high-quality technical delivery.
Engineering	Flutter Developer	Anita Jobby	Implements mobile app features using Flutter for both Android and iOS platforms.
Engineering / Security	Application Security Engineer	Eggs Sausted	Implements application-level security controls and addresses security vulnerabilities.
Engineering	Backend Developer	Mike Rophone	Develops backend logic and supports data handling functionality.
Engineering	Backend & Database Engineer	James Baxter	Designs and maintains local database structures and CRUD operations.
Quality Assurance	QA Engineer	Paige Turner	Executes test cases, logs bugs, and verifies fixes during development cycles.
Operations / Release	Operations & Release Manager	Barry D. Hatchet	Coordinates release planning, deployment readiness, and operational handover.
Operations / Release	DevOps Engineer	Eileen Sideways	Manages deployment pipelines and supports stable build and release processes.
Operations / Release	Quality Assurance Analyst	Kathryn Stockett	Supports QA documentation, defect tracking, and quality reporting.
Operations / Support	Customer Support Specialist	Harry Upman	Handles user issues, support tickets, and escalates problems to technical teams.
Enterprise	Enterprise	Thaedeus Tho	Oversees enterprise rollout and

	Manager		coordinates enterprise-level delivery requirements.
Enterprise	Enterprise Solutions Engineer	Tze Han	Supports enterprise deployments and resolves client-specific technical issues.
Enterprise	Release & Deployment Specialist	Alfred Yap	Executes deployment activities and coordinates release timelines.
Enterprise / Customer	Customer Success Executive	Jolie Ngai	Manages onboarding, training, and ongoing client success.
Enterprise / Operations	Operations Support Analyst	Koh Tong Wei	Monitors operational issues and supports day-to-day enterprise operations.

8.6 Budget Breakdown Expense Rationale

We chose this budget structure because it reflects a realistic end-to-end mobile app delivery workflow and allocates costs to the departments that actually own each responsibility. More funding is placed in Operations (COO) since the largest effort is building and integrating the core system (Flutter development, local database/CRUD, and QA across both iOS and Android). Product (CPO) is funded early to reduce rework by locking down requirements and validating UI/UX before heavy development starts. A dedicated Cybersecurity allocation under Finance/Risk (CFO) ensures protection of sensitive financial data (secure authentication and encrypted storage) is treated as a required component rather than an afterthought. Finally, a smaller Marketing (CMO) portion supports basic launch readiness (branding and promo materials) without inflating scope, since the project excludes ongoing marketing campaigns and paid subscriptions.

8.6.1 Finance Budget Breakdown Sources

<https://agileengine.com/software-development-cost-breakdown-in-2025-a-complete-guide/>

<https://biosistemika.com/blog/estimate-costs-of-software-development-projects/>

<https://clutch.co/agencies/app-marketing/pricing>

<https://www.businessofapps.com/ads/cpi/research/cost-per-install/>

9. Non-Plagiarism Declaration

We hereby declare that:

- We fully understand and agree to the abovementioned policy.
- We did not copy any materials from others or from other places.
- We did not share our materials with others or upload to any other places for public access.
- We agree that we will not disclose any information or material of the team project to others or upload to any other places for public access.
- We agree that our project will receive Zero mark if any misalignment with the above mentioned policies is detected.

Name of Team Members:

Name	Student ID
Tan Tze Han	2301483
Jolie Ngai Ning Li	2403175
Koh Tong Wei	2402162
Tho Jun En Thaedeus	2401039
Yap Hong Xiang Alfred	2401438